

Apollo Diagnostics - Laboratory Investigation Report

123, Nungambakkam High Road, Chennai - 600034 | NABL Accredited | Tel: 044-2827-5000

Patient Name:	Murugavel P	Report ID:	APL-ANAE-45-MUR-2026
Age / Gender:	45 Yrs / Male	Category:	B12 / Folate Deficiency Anaemia
Weight / Height:	68 kg / 170 cm	BMI:	23.5 kg/m2
Ref. Doctor:	Dr. Vikram Rao	Report Date:	02-Feb-2026

1. COMPLETE BLOOD COUNT [Circulatory System]

Red Cell Indices

Test	Result	Unit	Reference Range	Flag
Haemoglobin (Hb)	8.9	g/dL	13.0 - 17.5	LOW
Packed Cell Volume (PCV)	34.1	%	40 - 52	LOW
MCV	110.9	fL	83 - 101	HIGH
MCH	36.7	pg	27 - 32	HIGH
RDW-CV	14.7	%	11.6 - 14.0	HIGH

Haematology Comment: Haemoglobin 8.9 g/dL - macrocytic anaemia (MCV 110.9 fL). Consider B12/Folate deficiency or hypothyroidism. Peripheral smear and B12/Folate levels advised.

White Cells & Platelets

Test	Result	Unit	Reference Range	Flag
Total WBC	7000	cells/cmm	4500 - 11000	Normal
Neutrophils	62	%	40 - 75	Normal
Lymphocytes	28	%	20 - 45	Normal
Platelet Count	2.8	lakh/cmm	1.5 - 4.0	Normal

Comment: WBC and platelet counts within reference limits. No significant leucocytosis, leucopenia or thrombocytopenia.

2. IRON STUDIES [Haematopoietic System]

Test	Result	Unit	Reference Range	Flag
Serum Iron	98	ug/dL	60 - 170	Normal
TIBC	310	ug/dL	250 - 370	Normal
Transferrin Saturation	35.0	%	20 - 50	Normal
Serum Ferritin	55.1	ng/mL	30 - 300	Normal

Iron Studies Comment: Iron stores and transport markers are within normal limits. No iron deficiency or iron overload.

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3. HEPATIC FUNCTION TEST [Hepatobiliary System]

Liver Enzymes

Test	Result	Unit	Reference Range	Flag
SGPT / ALT	22	U/L	7 - 56	Normal
SGOT / AST	20	U/L	10 - 40	Normal
Alkaline Phosphatase (ALP)	70	U/L	44 - 147	Normal
Total Bilirubin	0.6	mg/dL	0.2 - 1.2	Normal
Albumin	4.2	g/dL	3.5 - 5.0	Normal

Hepatologist Note: Liver enzymes within normal limits. No hepatocellular damage or cholestatic process detected.

4. RENAL FUNCTION TEST [Urinary System]

Test	Result	Unit	Reference Range	Flag
Serum Creatinine	0.9	mg/dL	0.7 - 1.3	Normal
Blood Urea Nitrogen	14	mg/dL	7 - 20	Normal
eGFR (CKD-EPI)	98	mL/min/1.73m2	60 - 120	Normal
Serum Uric Acid	4.8	mg/dL	2.4 - 7.0	Normal
Sodium (Na+)	139	mEq/L	135 - 145	Normal
Potassium (K+)	4.0	mEq/L	3.5 - 5.1	Normal

Nephrologist Note: eGFR 98 mL/min/1.73m2 - glomerular filtration well preserved. Renal function normal.

5. LIPID PROFILE [Cardiovascular Risk]

Test	Result	Unit	Reference Range	Flag
Total Cholesterol	172	mg/dL	0 - 200	Normal
HDL Cholesterol	55	mg/dL	40 - 999	Normal
LDL Cholesterol	98	mg/dL	0 - 100	Normal
Triglycerides	88	mg/dL	0 - 150	Normal

Cardiologist Note: Lipid profile within optimal range. Cardiovascular risk from dyslipidaemia is low.

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6. THYROID FUNCTION TEST [Endocrine System]

Test	Result	Unit	Reference Range	Flag
TSH	2.1	uIU/mL	0.4 - 4.0	Normal
Free T3	3.2	pg/mL	2.3 - 4.2	Normal
Free T4	1.1	ng/dL	0.8 - 1.8	Normal

Endocrinologist Comment: Thyroid function normal. TSH 2.1 uIU/mL. Euthyroid state confirmed.

7. DIABETES SCREENING [Endocrine - Pancreas]

Test	Result	Unit	Reference Range	Flag
Fasting Blood Glucose	88	mg/dL	70 - 100	Normal
Post-Prandial Glucose (2hr)	115	mg/dL	0 - 140	Normal
HbA1c	5.1	%	0 - 5.7	Normal
HOMA-IR	1.5		0 - 2.5	Normal

Endocrinology Note: Glucose metabolism normal. HbA1c 5.1% - no diabetes or pre-diabetes. HOMA-IR 1.5 - no insulin resistance.

8. VITAMINS & BONE [Nutritional / Skeletal]

Test	Result	Unit	Reference Range	Flag
Vitamin D (25-OH)	38.0	ng/mL	30 - 100	Normal
Vitamin B12	137	pg/mL	211 - 911	LOW
Folate (Serum)	2.6	ng/mL	5.4 - 999	LOW
Calcium (Total)	9.4	mg/dL	8.5 - 10.5	Normal
PTH	32	pg/mL	15 - 65	Normal

B12 Note: Vitamin B12 137 pg/mL is low. Risk of megaloblastic anaemia and peripheral neuropathy. Intramuscular B12 injections (1000 mcg monthly) or high-dose oral supplementation (1000 mcg/day) recommended.

9. INFLAMMATORY MARKERS [Immune System]

Test	Result	Unit	Reference Range	Flag
ESR	25	mm/hr	0 - 15	HIGH
CRP	4.5	mg/L	0 - 5	Normal
hs-CRP	0.6	mg/L	0 - 1.0	Normal

Immunology Comment: CRP 4.5 mg/L and ESR 25 mm/hr are elevated. hs-CRP 0.6 mg/L - low. Inflammatory markers elevated - correlate with clinical symptoms and autoimmune screen.

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CLINICAL SUMMARY & RECOMMENDATIONS

- [ABNORMAL]** Anaemia: Hb=8.9 g/dL (LOW). Macrocytic pattern - check B12/Folate.
- [DEFICIENT]** Vitamin B12=137 pg/mL (LOW). B12 supplementation required.
- [ACTION]** Next appointment: Dr. Vikram Rao in 4-8 weeks for follow-up and review of supplementation response.

Report reviewed by: Dr. Anitha Krishnamurthy, MD (Pathology) | Verified by Dr. Vikram Rao
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